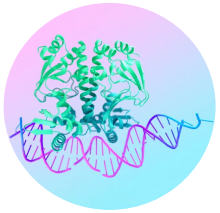
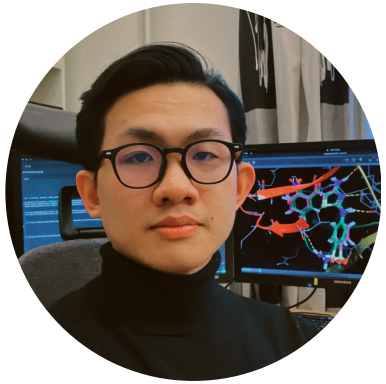
SciLifeLab

AI agent collaboration with the Model Context Protocol (MCP)





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What you will build today...

- ❑ A drug discovery **MCP server** exposing tools
- ❑ A **MCP client** that discovers and calls tools dynamically
- ❑ **Security**: detect and defend against tool poisoning attacks
- ❑ **Guardrails**: input validation, blocked compounds, rate limiting
- ❑ **Integration**: connect your earlier **LangGraph agent** to MCP
- ❑ **Bonus 🎁: MCP Python SDK,** from basic to production
- ❑ **Bonus 🎁: integrating with real-world research works**!

Outline: The bigger picture

SESSION 2 — BUILDING THE MCP PIPELINE

📐

1

Raw Protocol

Understand MCP from scratch: JSON-RPC, resources, tools

Parts 1 – 2

READING

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🔧

2

Working Server

Build your own drug discovery MCP server & client

Part 3

HANDS-ON

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3

Safe Server

Add guardrails, security layer, and tool poisoning defence

Parts 4 – 5

HANDS-ON

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4

Connected Agent

LangGraph + MCP working together end-to-end

Part 6

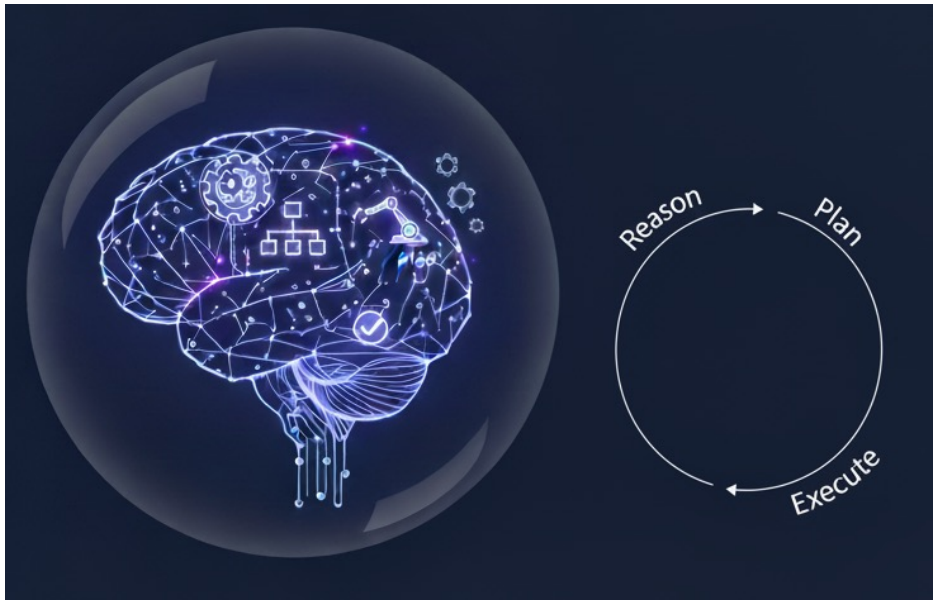
HANDS-ON

By the end of this session, you will have built every stage of this diagram.

[1]

Our journey so far (session 1 recap)

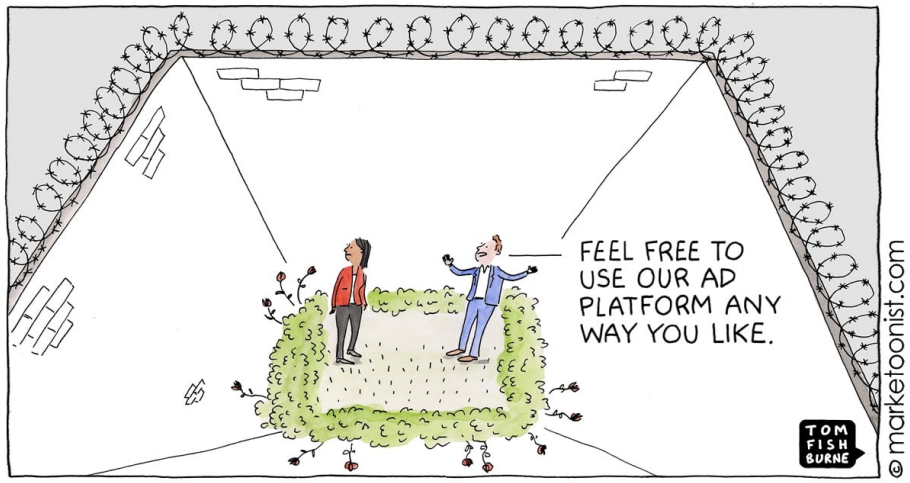
- In Session 1 you built a **LangGraph** drug discovery agent
- Three tools: **`resolve_smiles`**, **`get_properties`**, **`lit_search`**
- It can **reason, plan, and execute** a sequence of actions
- A working proof-of-concept for agentic AI in life sciences



[2]

But it was a walled garden!

- Your tools were **hardcoded** inside your Python file
- `resolve_smiles`, `get_properties`, and `lit_search` **only worked in your notebook**
- Any other agent or researcher had to **copy** your code **to use** them
- Perfect for a demo, but not for a real research environment



[3]

The Real-world challenges: scalability & reinventing the wheel

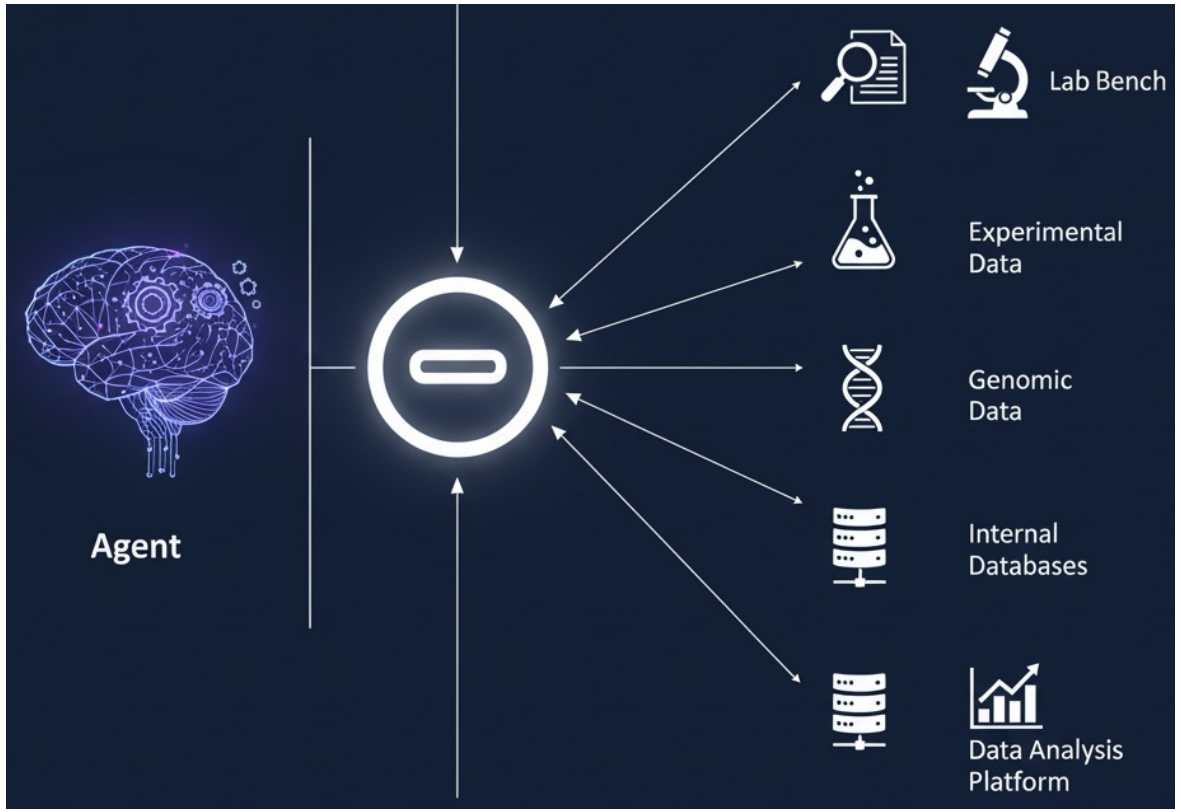
[2]

The Real-world challenges: scalability & reinventing the wheel

- Need a **new tool** (e.g. ChEMBL, a private drug database)? → **Stop, code, redeploy**
- **Not flexible enough** for fast-paced drug discovery
- Every team builds **isolated agents** and **duplicates** effort
- Teams build their own agents and tools **independently**
- **Duplication, inconsistency, maintenance** burden
- **No standard way to share** or discover tools across groups

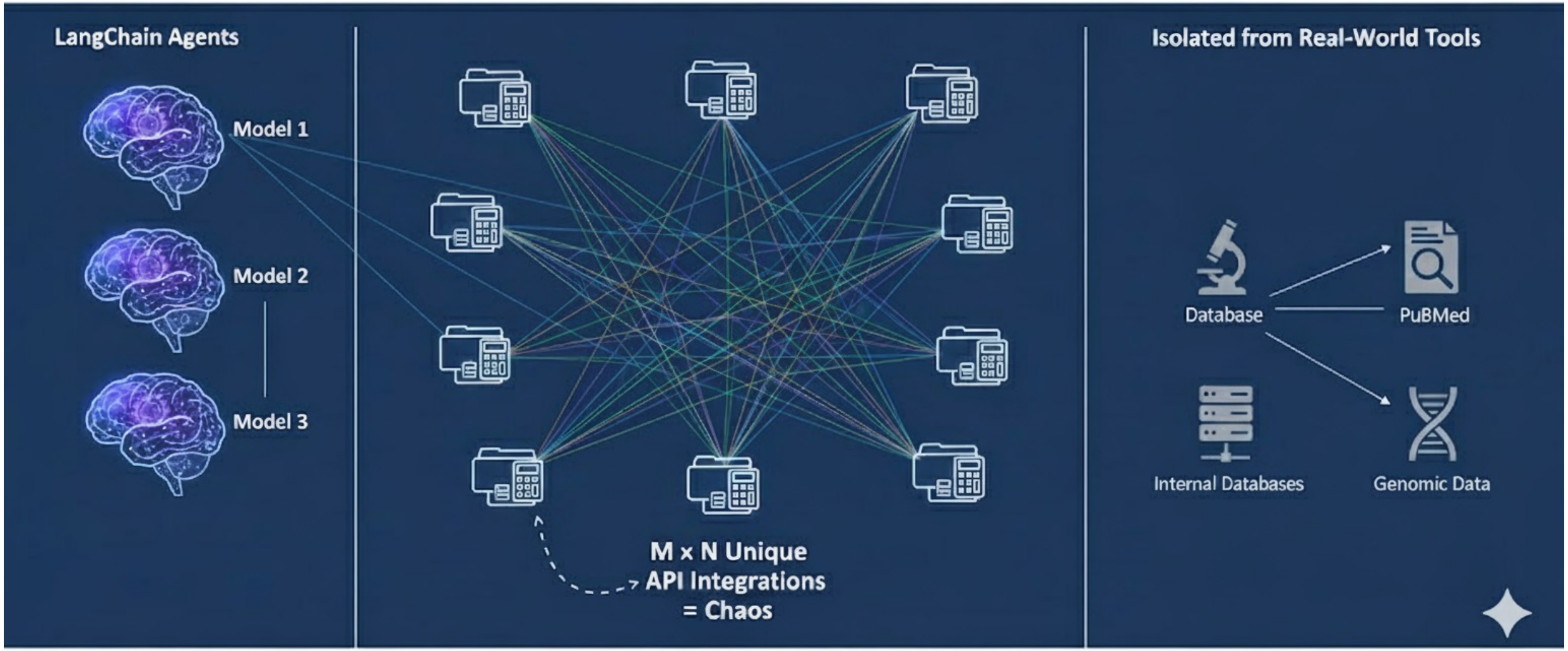
Solution: introducing MCP (Model Context Protocol)!

- LangGraph = **The brain** (reasoning and planning)
- MCP = **The nervous system** (standardised connections)
- **Dynamic tool discovery**: agents find and use tools automatically
- **Unified ecosystem**: teams build and share tools composably



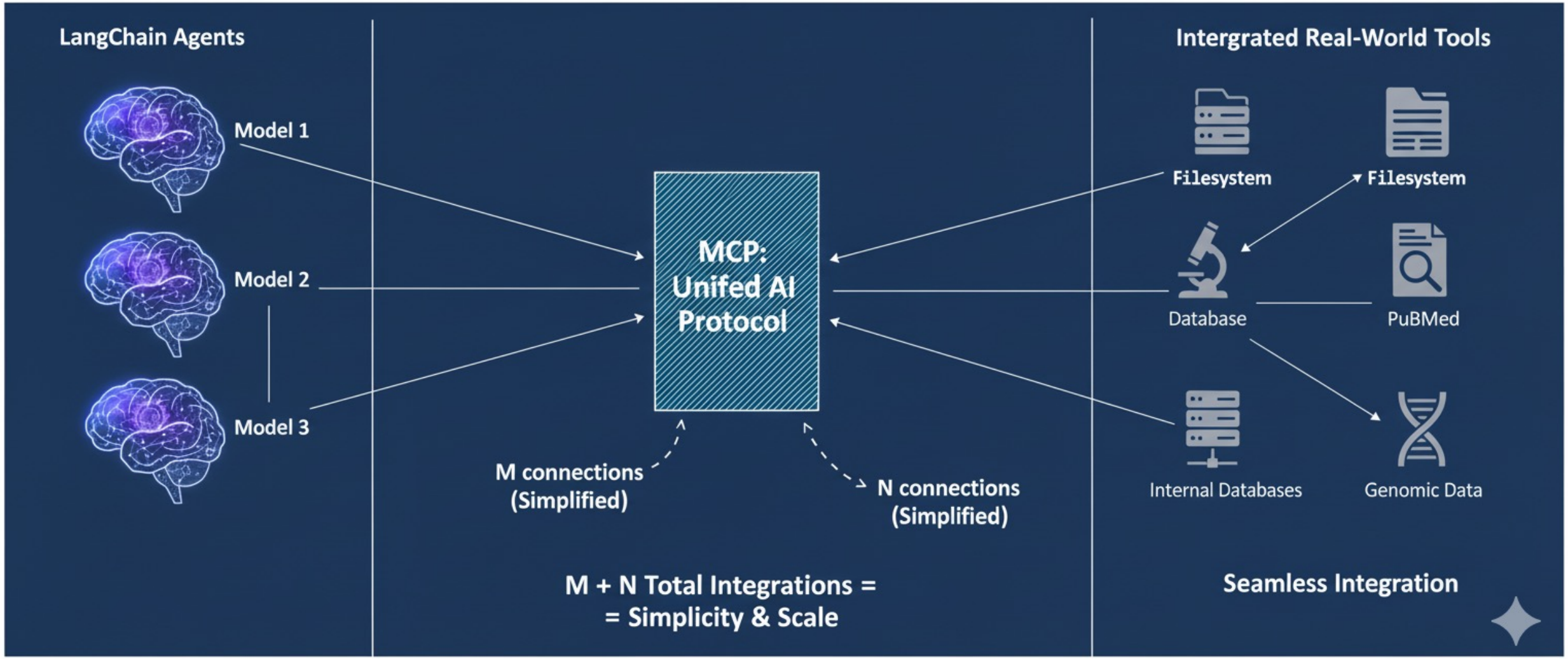
MCP: The **"USB-C for AI"** [2]

The M×N integration problem



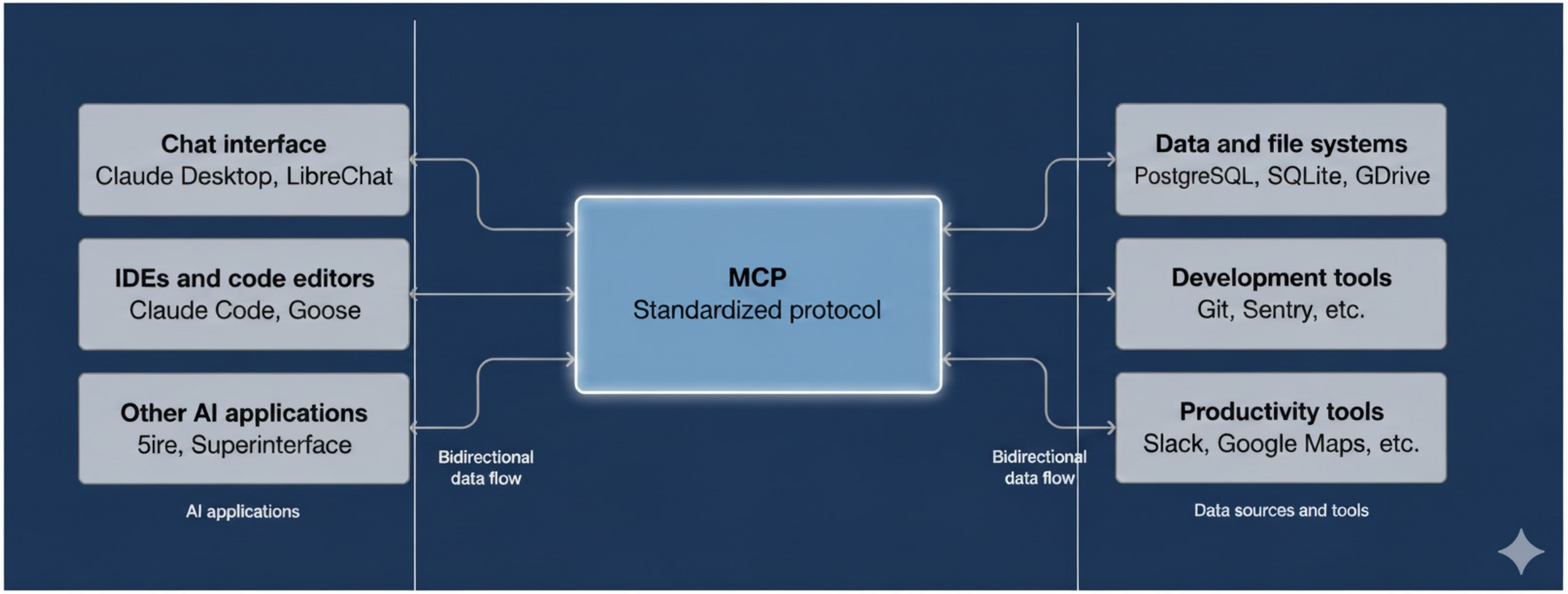
[4]

With MCP (M+N solution)



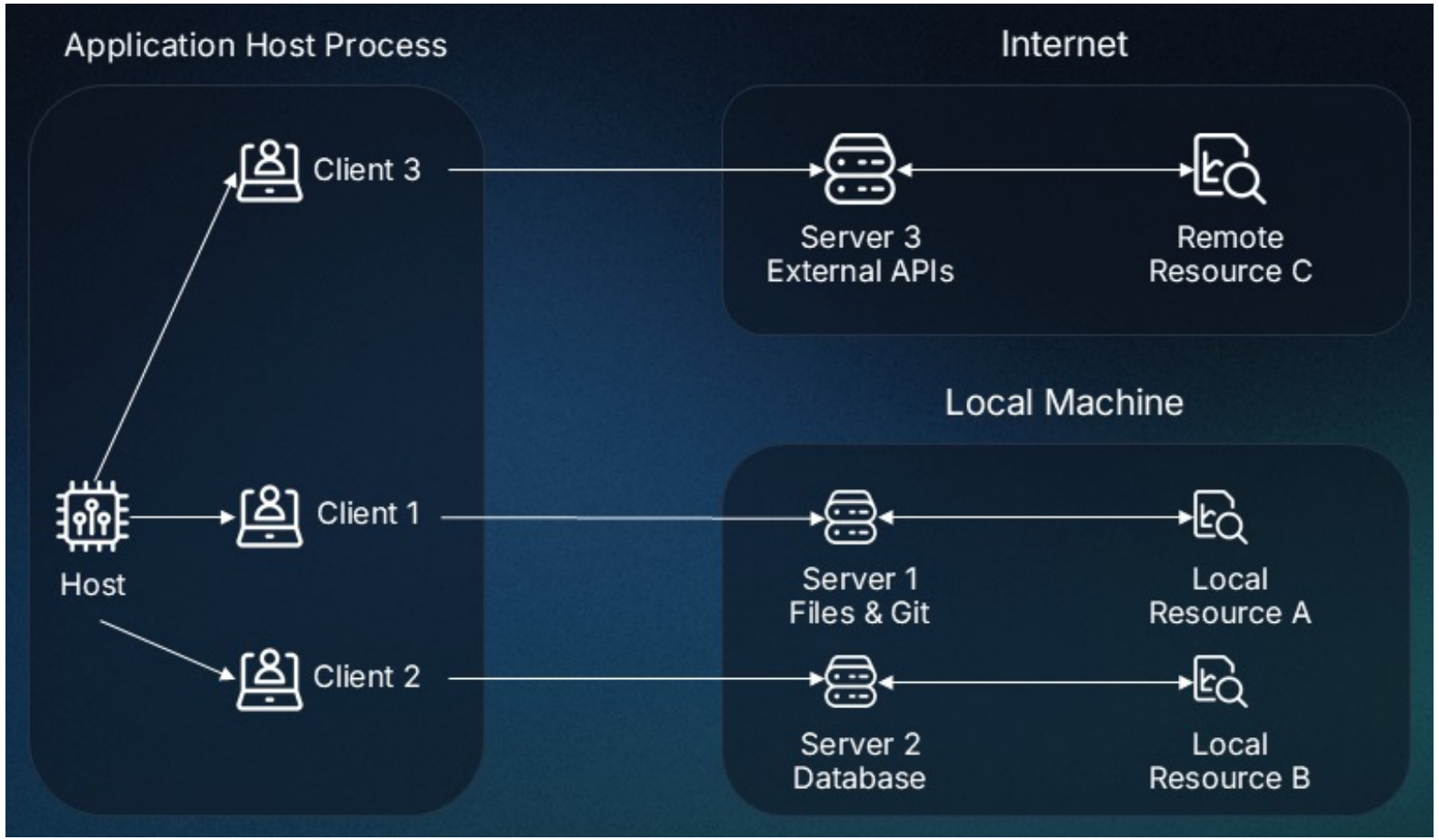
[4]

MCP: overview



[5]

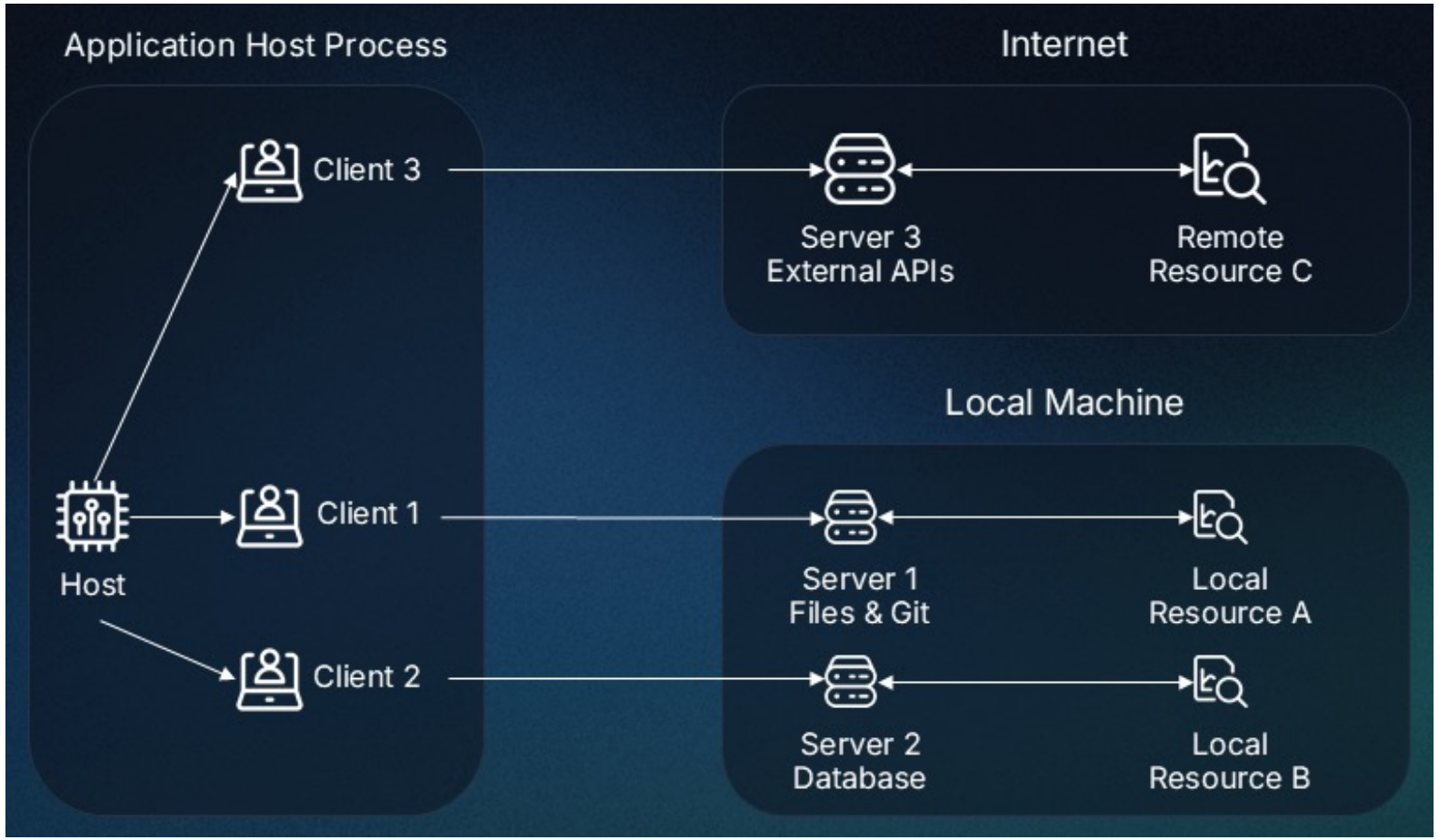
MCP components: Host



[6]

Host: User-facing AI app that interacts with end-users and orchestrates requests, LLM processing, and external tools.

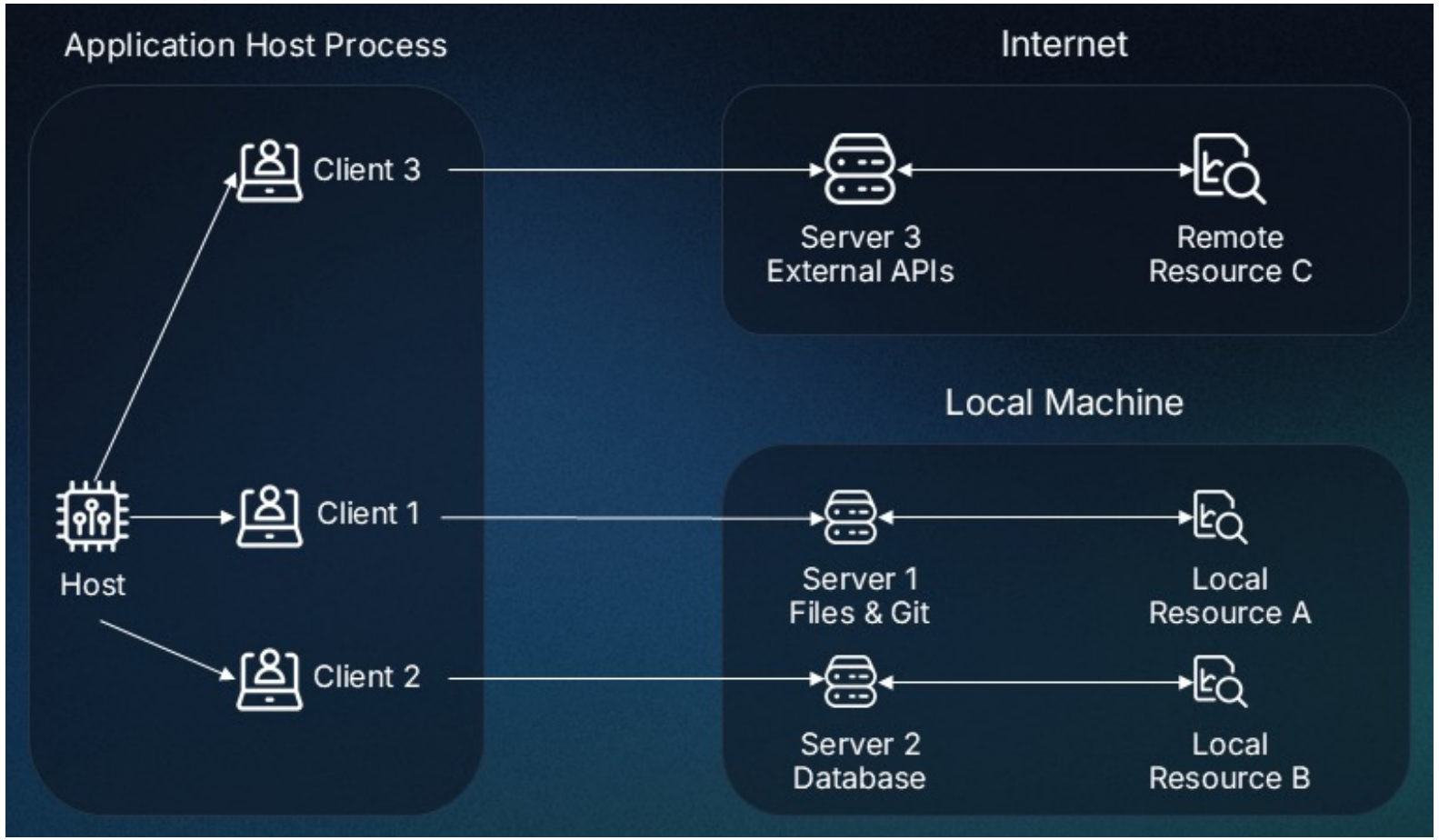
MCP components: Client



[6]

Client: Manages communication between the Host and a single MCP Server, handling protocol details.

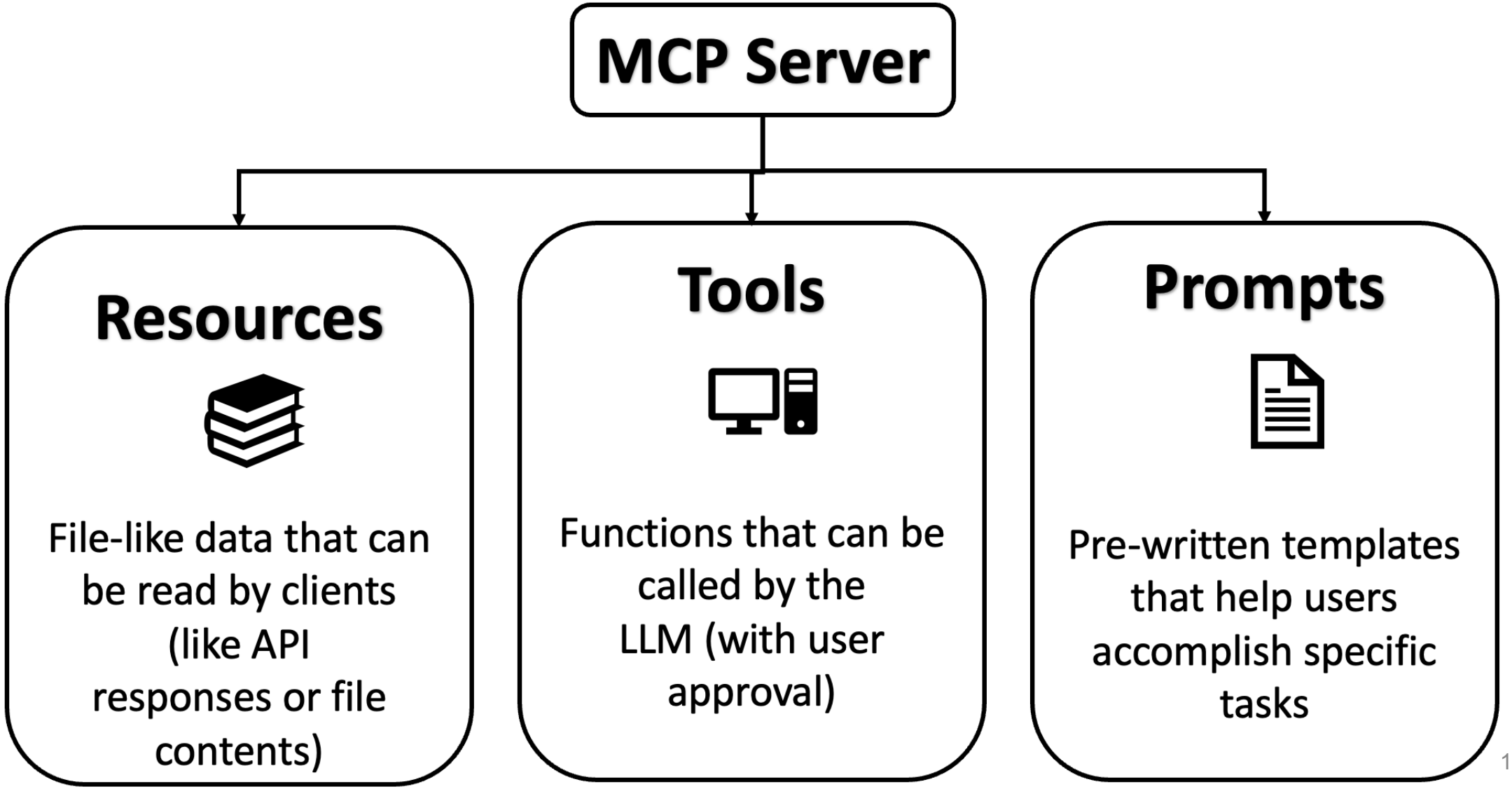
MCP components: Server



[6]

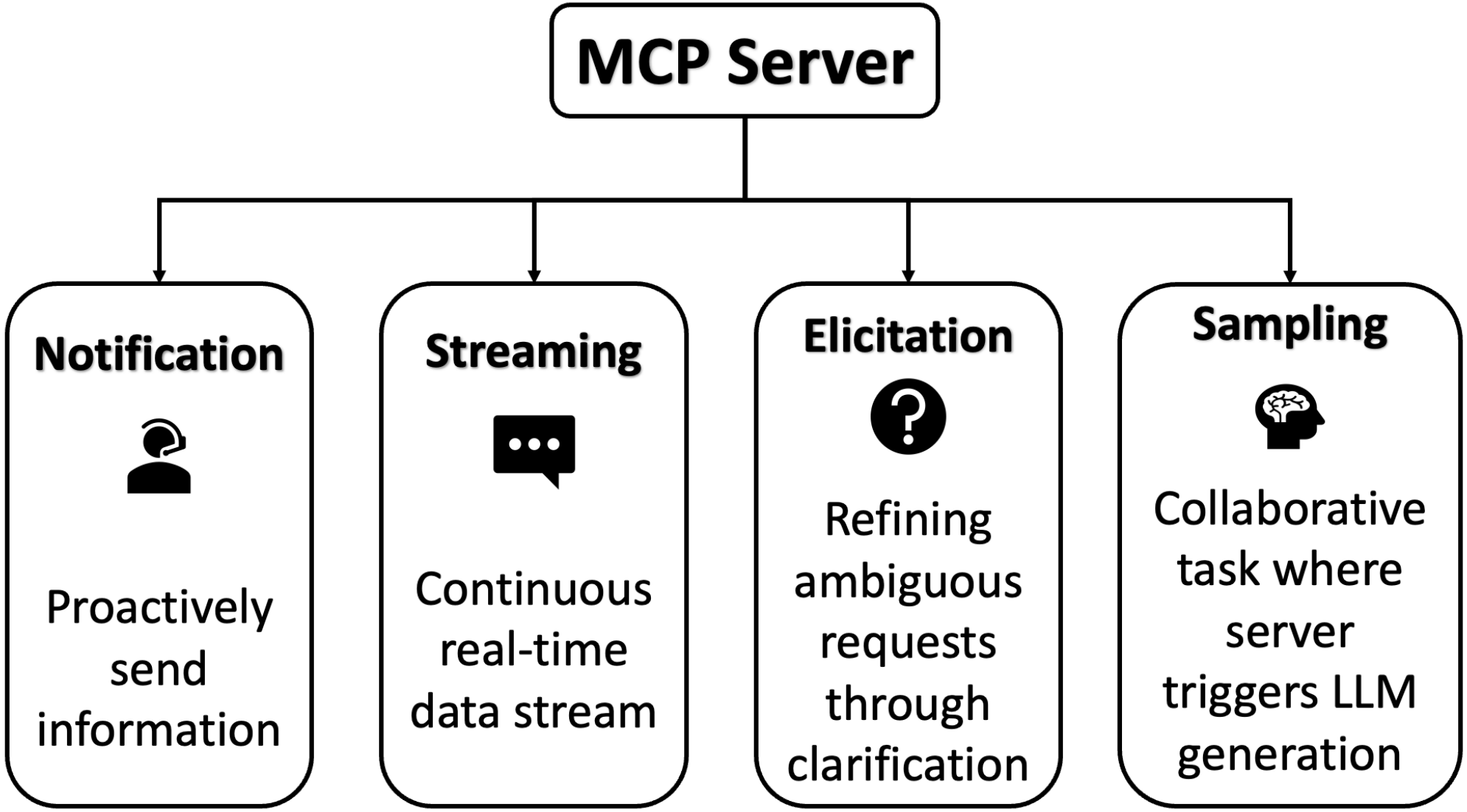
Server: External service exposing tools, resources, or prompts via the MCP protocol.

MCP capabilities: basic



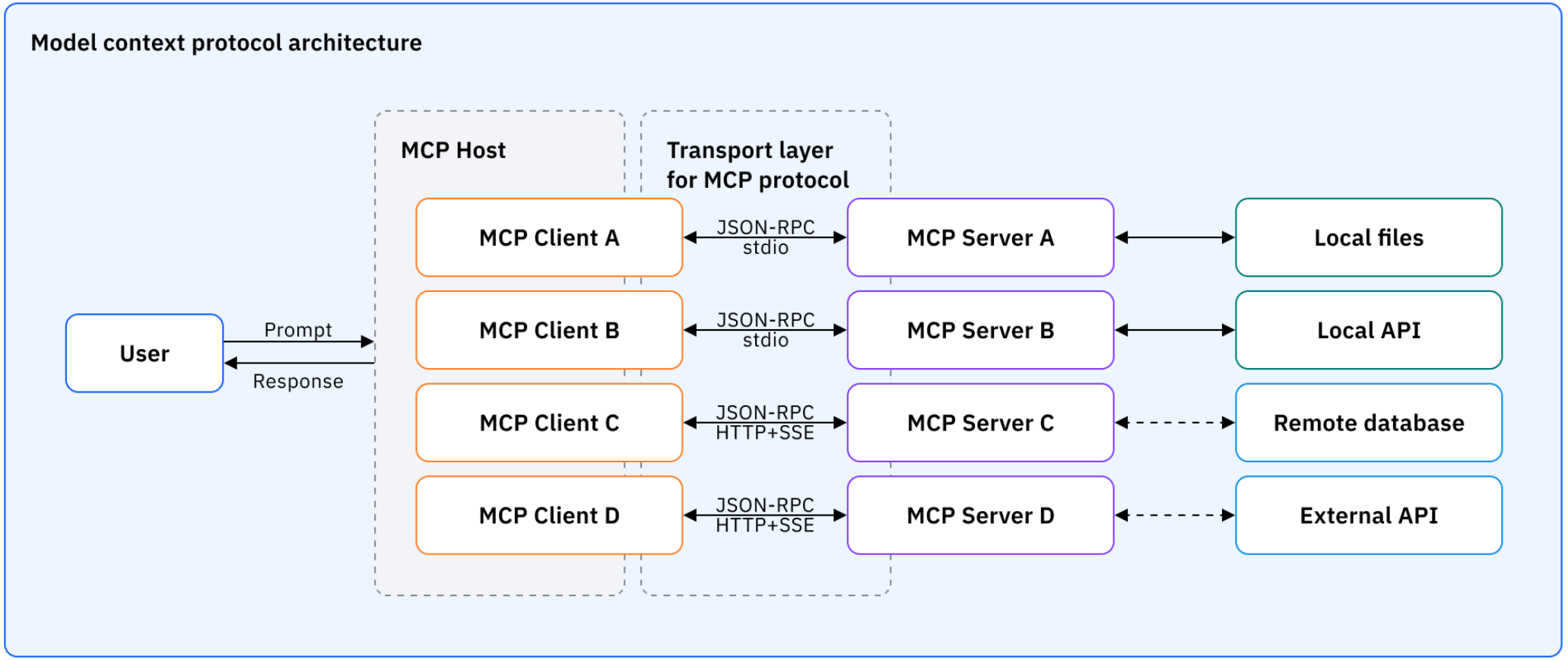
[7]

MCP capabilities: advanced



[7]

MCP layers: data layer



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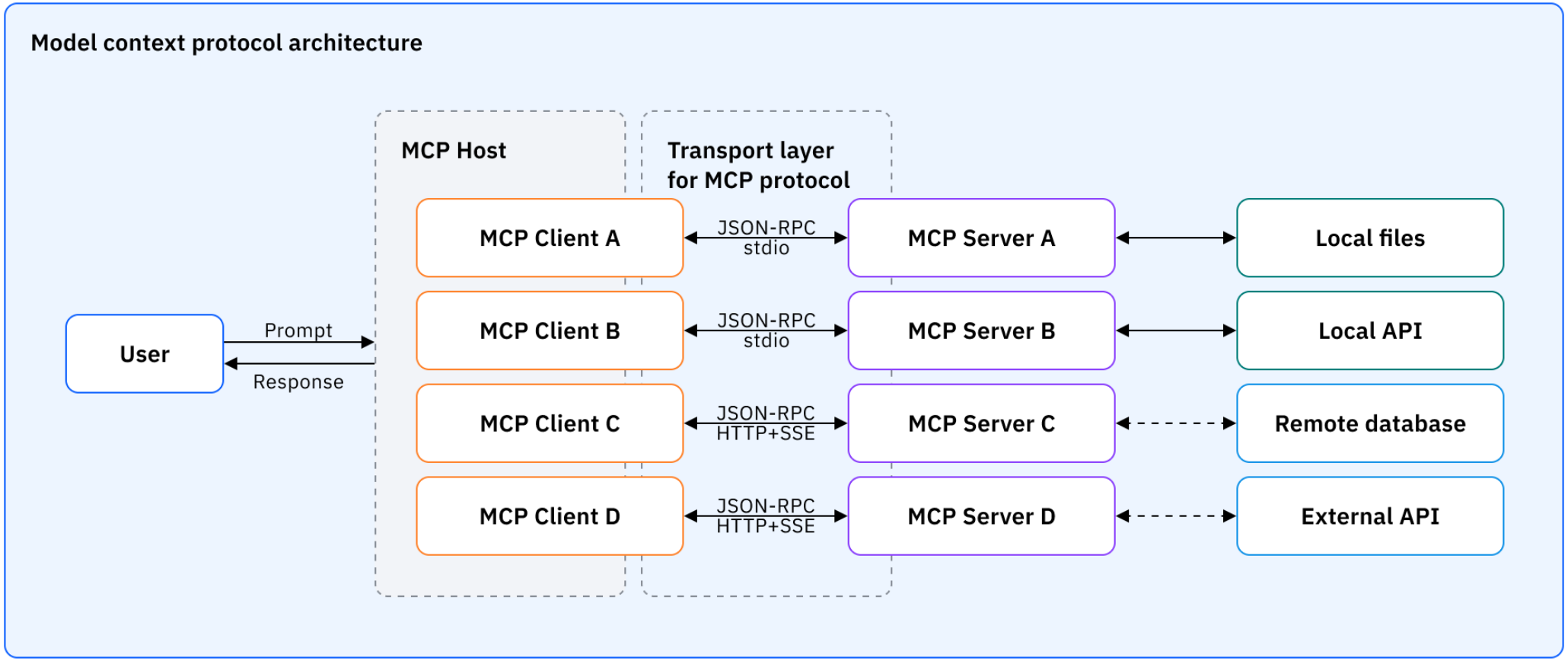
Data layer: JSON-RPC protocol defining lifecycle management, tools, resources, prompts, and notifications.

MCP layers: transport layer



[8]

Transport layer: Communication channels, connection handling, message framing, and authentication.

Why security matters in MCP

- Tool descriptions are read by the LLM and **directly influence** its decisions
- In an open MCP ecosystem you may connect to servers **you didn't write**
- A crafted description is indistinguishable from a legitimate one at the protocol level, this is called **tool poisoning**

MCP Security Hands On: Tool poisoning & Guardrails

- In **Part 5.3** you will build this **tool poisoning** attack and then defend against it

- **Guardrails** you will implement:
  - ✓ SMILES validation
  - ✓ blocked compounds
  - ✓ rate limiting
  - ✓ description sanitisation

Bonus 🎁 MCP Python SDK: from basic to production

Zero boilerplate: decorators replace manual JSON-RPC handling

One-line server startup: mcp.run() vs. custom HTTP/SSE plumbing

Built-in LangChain bridge: no adapter code required

Type-safe by default: Pydantic validation out of the box

SDK speed: same drug discovery agent, faster implementation

Bonus 🎁 : SciLifeLab Serve Public App + MCP !

FROM WORKSHOP TO REAL WORLD

🧠 LangGraph Agent

Host: reasoning & planning

① **User query** — natural language

② **Mediator** — route to MCP tool

③ **Tool call** — via MCP protocol

④ **Diagnosis** — grounded output

Same pattern from Session 1 + Part 6

JSON-RPC →

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🔌 MCP Server

Flask: validate, parse, guardrails

tools/list
→ analyze_xray schema

tools/call
→ validate input + forward to Gradio

Guardrails
✓ Image validation ✓ Rate limiting
✓ Format check ✓ Timeout (120s)

Response
→ probabilities + heatmap paths

Same pattern from Parts 3 + 5

HTTPS →

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☁️ SciLifeLab Serve

Any deployed public app

SHAMSUL

Chest X-ray pathology + Interpretability analysis
Chest X-ray → prediction probabilities + heatmaps
(Grad-CAM · LIME · SHAP · LRP)

🧬 Protein structure prediction

🧪 Chemical structure recognition

🔬 Any public app with exposed APIs

Free for Swedish research institutions

Same MCP pattern you built today — works with any app on SciLifeLab Serve

serve.scilifelab.se/apps · Bonus_Gradio_MCP/ in the workshop repo

[1]

Additional resources

Official MCP documentation

https://modelcontextprotocol.io/docs/getting-started/intro

A nice blog post

https://www.ramotion.com/blog/mcp-protocol/

Hugging Face course

https://huggingface.co/learn/mcp-course/en/unit1/introduction

References

[1] - AI-generated by Mahbub using Claude Code, Opus 4.6 Extended

[2] - AI-generated by Mahbub using Google Gemini 2.5 Pro

[3] - https://marketoonist.com/2022/05/walled-gardens.html

[4] - AI-generated by Mahbub using Google Gemini 2.5 Pro.
Original idea adapted from https://huggingface.co/learn/mcp-course/en/unit1/key-concepts

[5] - https://modelcontextprotocol.io/docs/getting-started/intro

[6] - https://medium.com/@sathishsuraj/understanding-the-model-context-protocol-mcp-architecture-implementation-and-use-cases-499f1e72673b

[7] - Original idea adapted from https://www.inferless.com/learn/how-to-connect-everyday-tools-with-mcp

[8] - https://www.ibm.com/think/podcasts/mixture-of-experts/ai-action-plan-chatgpt-agents-deepmind-imo

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HANDS-ON

You have just built every stage of this diagram. The walled garden is open.

[1]

Next steps

shorturl.at/TLx0j

Scan or use this link!

- Make sure your **OpenAI API key is ready**
- Start and explore! ☺